KSA Office IT Support & Infrastructure Management

Phase 07: IT Support Operations

Afaq Business Solutions Co.

Service & Systemd Troubleshooting SOP

1. Purpose

Provide a controlled procedure for investigating Linux service availability issues using systemd status information and service logs, applying safe remediation, and verifying service operation.

2. Workflow

- Document the reported service availability issue and create the support ticket.
- Identify an appropriate service for investigation.
- Review service status with systemctl status.
- Review recent service events with journalctl.
- Determine whether the service is stopped, failed, or unavailable.
- Avoid unrelated system configuration changes.
- Restart the affected non-critical or authorized service when appropriate.
- Verify service status after remediation.
- Review recent logs to confirm normal operation.
- Document the investigation and resolution and close the ticket.

3. Key Commands

- **systemctl list-units --type=service --state=running** - list running services.
- **systemctl status <service-name> --no-pager** - inspect service state.
- **journalctl -u <service-name> -n 20 --no-pager** - review recent service logs.
- **sudo systemctl restart <service-name>** - restart an authorized service.
- **systemctl status <service-name> --no-pager** - verify service operation.

4. Safe Remediation

- Use a non-critical or explicitly authorized service for training.
- Do not stop or restart essential production services for practice.

- Make the smallest change necessary.
- Confirm the service is operational after the change.

5. Verification

- Confirm the service reports the expected active state.
- Review recent logs for errors or abnormal termination.
- Test relevant functionality when safely possible.
- Record the verification result before closing the ticket.

6. Evidence & Security

- Capture only evidence necessary for the incident.
- Do not expose credentials, tokens, or unrelated sensitive system information.
- Use sanitized or controlled lab services for screenshots.

7. Best Practices

- Inspect before changing.
- Use logs to support diagnosis.
- Apply minimal, controlled remediation.
- Verify after corrective actions.
- Maintain clear incident documentation.